piloting, extraterrestrial drones must rely on autonomous navigation and onboard decision algorithms, allowing them to adapt to terrain changes, weather variations, and unforeseen hazards independently.

By extending the reach of exploration missions, drones can greatly increase scientific return, enabling researchers to develop a more complete understanding of planetary histories, potential resources, and suitability for future human habitation. Thus, drones are not merely supporting tools, but critical scientific platforms in the expansion of humanity's presence beyond Earth.

9.7.2 Developing drones for missions on Mars

Designing drones for extraterrestrial missions requires overcoming environmental, mechanical, and operational challenges that differ significantly from conditions on Earth. Each planetary body presents unique constraints such as atmospheric density, gravity levels, surface composition, temperature extremes, and radiation exposure that directly influence drone structure, propulsion, and control systems. For example, Mars has a thin atmosphere, roughly 1% the density of Earth's, which reduces aerodynamic lift and requires drones to use large, lightweight rotor blades, high rotational speeds, and highly efficient power systems to maintain flight. NASA's Ingenuity Mars Helicopter demonstrated that even under such conditions, controlled flight is possible when drones are ultra-lightweight and supported by strong autonomous navigation algorithms.

On the Moon, the absence of an atmosphere eliminates the possibility of aerodynamic lift entirely, prompting the development of propulsion-based "hopper" drones that use small rocket thrusters or compressed gas bursts to make controlled jumps across the surface. These systems must be engineered to withstand extreme temperature fluctuations, abrasive lunar dust, and low-gravity dynamics, where even minimal thrust can create large